An Example: 2D Heat Equation

1 Model Problem

Consider the 2D heat equation on the space-time cylinder $Q := \Omega \times (0,T)$:

$$\partial_t u - \Delta u = f \quad \text{in } Q,$$

with homogeneous Dirichlet boundary conditions

$$u = 0 \quad \text{on } \partial\Omega \times (0,T),$$

and initial condition

$$u(x,0) = u_0.$$

2 Weak Formulation

Multiply the PDE by a test function v and integrate over Ω:

$$\langle \partial_t u, v \rangle + \langle \nabla u, \nabla v \rangle = \langle f, v \rangle \qquad \forall v \in H_0^1(\Omega).$$

In space-time form, define the spaces

$$U := \left\{ u \in L^2(0,T;H_0^1(\Omega)) \; : \; \partial_t u \in L^2(0,T;H^{-1}(\Omega)) \right\}, \qquad V := L^2(0,T;H_0^1(\Omega)).$$

Then the weak problem can be written abstractly as: Find $u \in U$ such that

$$A(u)(v) = 0 \quad \forall v \in V,$$

where

$$A(u)(v) := \int_0^T \Big(\langle \partial_t u, v \rangle + \langle \nabla u, \nabla v \rangle - \langle f, v \rangle \Big) \, \mathrm{d}t.$$

3 Space-time Finite Element Discretization

Let $U_h \subset U$ and $V_h \subset V$ be finite-dimensional subspaces. The discrete primal problem is:

$$\text{Find } u_h \in U_h \text{ such that } A(u_h)(v_h) = 0 \qquad \forall v_h \in V_h.$$

For the discrete adjoint, one solves

$$\text{Find } z_h \in V_h \text{ such that } A'(u_h)(v_h, z_h) = J'(u_h)(v_h) \qquad \forall v_h \in U_h.$$

4 Goal Functionals

Typical goal functionals include:

$$J(u) = \int_\omega u(x, T)\, \mathrm{d}x,$$

where $\omega \subset \Omega$ is a target region.

5 DWR Adjoint Problem

The adjoint problem is defined by

$$A'(u)(v, z) = J'(u)(v) \qquad \forall v \in U.$$

The unknown is the adjoint state z.

5.1 Gâteaux Derivative

For a direction v, the Gâteaux derivative of A is

$$A'(u)(v, z) = \lim_{\varepsilon \to 0} \frac{A(u + \varepsilon v)(z) - A(u)(z)}{\varepsilon}.$$

For the heat equation this gives

$$A'(u)(v, z) = \int_0^T \Big(\langle \partial_t v, z \rangle + \langle \nabla v, \nabla z \rangle \Big)\, \mathrm{d}t.$$

5.2 Integration by Parts

Using integration by parts in time, the adjoint weak form becomes

$$\langle v(T), z(T)\rangle + \int_0^T \langle v, -\partial_t z - \Delta z\rangle \, \mathrm{d}t = J'(u)(v) = \langle v(T), \mathbf{1}_\omega\rangle \, .$$

5.3 Adjoint Strong Form

The adjoint strong form is

$$-\partial_t z - \Delta z = 0 \quad \text{in } \Omega \times (0, T),$$

$$z = 0 \quad \text{on } \partial\Omega \times (0, T),$$

$$z(\cdot, T) = \mathbf{1}_\omega.$$

6 Error Identity

From Theorem 1, let $\tilde{u} \in U_h$ and $\tilde{z} \in V_h$ be arbitrary approximations. Then the DWR error identity is

$$J(u) - J(\tilde{u}) = \frac{1}{2}\Big(\rho(\tilde{u})(z - \tilde{z}) + \rho^*(\tilde{u}, \tilde{z})(u - \tilde{u})\Big) - \rho(\tilde{u})(\tilde{z}) + R^{(3)},$$

where

$$\rho(\tilde{u})(\cdot) := -A(\tilde{u})(\cdot), \qquad \rho^*(\tilde{u}, \tilde{z})(\cdot) := J'(\tilde{u})(\cdot) - A'(\tilde{u})(\cdot, \tilde{z}).$$

A third-order remainder term is

$$R^{(3)} = \frac{1}{2}\int_0^1 \Big[J'''(\tilde{u}+se)(e,e,e) - A'''(\tilde{u}+se)(e,e,e,\tilde{z}+se^*) - 3A''(\tilde{u}+se)(e,e,e^*)\Big] s(s-1)\, \mathrm{d}s,$$

with

$$e = u - \tilde{u}, \qquad e^* = z - \tilde{z}.$$

7 Enriched Space Approximation

Let $U_h^{(2)}$ and $V_h^{(2)}$ be enriched spaces with

$$U_h \subset U_h^{(2)} \subset U, \qquad V_h \subset V_h^{(2)} \subset V.$$

The enriched primal and adjoint problems are

$$\text{Find } u_h^{(2)} \in U_h^{(2)} \text{ such that } A(u_h^{(2)})(v_h^{(2)}) = 0 \qquad \forall v_h^{(2)} \in V_h^{(2)},$$

$$\text{Find } z_h^{(2)} \in V_h^{(2)} \text{ such that } A'(u_h^{(2)})(v_h^{(2)}, z_h^{(2)}) = J'(u_h^{(2)})(v_h^{(2)}) \qquad \forall v_h^{(2)} \in U_h^{(2)}.$$

Then the computable estimator is

$$\eta^{(2)} = \frac{1}{2}\Big(\rho(\tilde{u})(z_h^{(2)} - \tilde{z}) + \rho^*(\tilde{u}, \tilde{z})(u_h^{(2)} - \tilde{u})\Big) - \rho(\tilde{u})(\tilde{z}) + R_h^{(3)}.$$

A useful splitting is

$$\eta^{(2)} = \underbrace{\frac{1}{2}\Big(\rho(\tilde{u})(z_h^{(2)} - \tilde{z}) + \rho^*(\tilde{u}, \tilde{z})(u_h^{(2)} - \tilde{u})\Big)}_{\eta_h^{(2)}} - \underbrace{\rho(\tilde{u})(\tilde{z})}_{\eta_k} + \underbrace{R_h^{(3)}}_{\eta_R^{(2)}}.$$